Continuous Convergence and Curried Functional Spaces

1. Preliminary definitions

1.1. The language of filters

Definition 1.1.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition 1.1.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ *generated* by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$.
If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x.
If $f : X \to Y$ is a map and $F \in \mathcal{F}(X)$ is a filter on X, then $\{f(A) \mid A \in F\}$ is a filter base that generates the *image filter* denoted $f(F)$.

Definition 1.1.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \to x$, if it contains all open neighborhoods of x.

Remark 1.1.1. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter
$$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$
converges to x. This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.
- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \varnothing$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X.
- A function $f : X \to Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \to x \in X$ implies $f(F) \to f(x) \in Y$.

Lemma 1.1.1. (see [1], page 74): *Let X be a set and $f_i : X \to Y_i$ be a family of maps, where Y_i are topological spaces. Let X be endowed with the coarsest topology such that f_i is continuous for each i. Then a filter $F \in \mathcal{F}(X)$ converges to $x \in X$ if and only if $f_i(F)$ converges to $f_i(x)$ for all i.*

1.2. The topology of pointwise convergence

Definition 1.2.1. (pointwise convergence topology, see [2]): Let X be a set, Y a topological space, and denote by $\mathcal{S}(X, Y)$ the set of functions from X to Y. For each $x \in X$ and each open set $U \subset Y$, let $V(x, U)$ be the set of all functions $f \in \mathcal{S}(X, Y)$ such that $f(x) \in U$. The collection

$$\{V(x, U) \mid x \in X, U \subset Y\}$$

forms a subbase of a topology on $\mathcal{S}(X, Y)$, called the *topology of pointwise convergence.*

Remark 1.2.1. Let X be a set and Y a topological space. It is not hard to see that a filter $F \in \mathcal{F}(\mathcal{S}(X, Y))$ converges to a function $f \in \mathcal{S}(X, Y)$ if and only if, for any $x \in X$, the filter $F(x) = [\{A(x) \mid A \in F\}]$ converges to $f(x) \in Y$.

Lemma 1.2.1. *Let X be a set and Y a Hausdorff topological space. Then $\mathcal{S}(X, Y)$ is also Hausdorff.*

Proof: Consider two functions $f, g \in \mathcal{S}(X, Y)$ such that $f \neq g$. This implies that there is $x_0 \in X$ such that $f(x_0) \neq g(x_0)$. Since Y is Hausdorff, there are neighborhoods $U_f \ni f(x_0)$ and $U_g \ni g(x_0)$ such that $U_f \cap U_g = \varnothing$. It then follows that $f \in V(x_0, U_f)$, $g \in V(x_0, U_g)$, and $V(x_0, U_f) \cap V(x_0, U_g) = \varnothing$. ■

Lemma 1.2.2. *Let X be a set and let Y be a real topological vector space. Then $\mathcal{S}(X, Y)$ is also a real topological vector space with respect to the topology of pointwise convergence and the pointwise linear structure.*

Proof: We need to show that the maps $+ : \mathcal{S}(X, Y) \times \mathcal{S}(X, Y) \to \mathcal{S}(X, Y)$ and $\cdot : \mathbb{R} \times \mathcal{S}(X, Y) \to \mathcal{S}(X, Y)$ are continuous.
To begin, consider two filters $F_1, F_2 \in \mathcal{F}(\mathcal{S}(X, Y))$, converging to f_1 and f_2 respectively. To show that $F_1 + F_2$ converges to $f_1 + f_2$, consider $x \in X$. We have

$$\begin{aligned} (F_1 + F_2)(x) &= [\{(A + B)(x) \mid A \in F_1, B \in F_2\}] \\ &= [\{A(x) + B(x) \mid A \in F_1, B \in F_2\}] \\ &= F_1(x) + F_2(x) \to f_1(x) + f_2(x) = (f_1 + f_2)(x). \end{aligned}$$

This shows that the addition operation is continuous. Scalar multiplication is continuous by a similar argument. ■

1.3. The compact-open topology

Definition 1.3.1. (compact-open topology, see [2]): Let X and Y be two topological spaces, and by $\mathcal{C}(X, Y)$ denote the set of all continuous functions from X to Y. For each compact set $K \subset X$ and each open set $U \subset Y$, let $V(K, U)$ be the set of all functions $f \in \mathcal{C}(X, Y)$ such that $f(K) \subset U$. The collection

$$\{V(K, U) \mid K \subset X, U \subset Y\}$$

forms a subbase of a topology on $\mathcal{C}(X, Y)$, called the *compact-open* topology.

Remark 1.3.1. Clearly, the topology of pointwise convergence on $\mathcal{C}(X, Y)$ is coarser than the compact-open topology.

Proposition 1.3.1. (see [3]): *Let X and Y be topological spaces such that X is locally compact and Y is regular. Then a filter $F \in \mathcal{F}(\mathcal{C}(X, Y))$ converges to a function $f \in \mathcal{C}(X, Y)$ if and only if, for any filter $P \in \mathcal{F}(X)$ converging to $p \in X$, we have $F(P) \to f(p)$ in Y, where the filter $F(P)$ is based on*

$$\{A(B) \mid A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \mid A \in F, B \in P\}.$$

Proposition 1.3.2. (see [2], page 302): *Let X, Y, Z be topological spaces such that X is Hausdorff and Y is locally compact. Then the map*

$$\gamma : \mathcal{C}(X \times Y, Z) \to \mathcal{C}(X, \mathcal{C}(Y, Z))$$

defined as $\gamma(f)(x)(y) = f(x, y)$, is well-defined and is a homeomorphism.

Lemma 1.3.1. *Let X, Y be topological spaces such that Y is Hausdorff. Then the space $\mathcal{C}(X, Y)$ is also Hausdorff.*

Proof: The compact-open topology on $\mathcal{C}(X, Y)$ is finer than the topology of pointwise convergence, and the latter is Hausdorff, which implies that the former is Hausdorff. ∎

Lemma 1.3.2. *Let X be a topological space and Y a real topological vector space. Then $\mathcal{C}(X, Y)$ is also a real topological vector space with respect to the pointwise linear structure and the compact-open topology.*

Proof: We will show that the map $+ : \mathcal{C}(X, Y) \times \mathcal{C}(X, Y) \to \mathcal{C}(X, Y)$ is continuous. Let $F_1, F_2 \in \mathcal{F}(\mathcal{C}(X, Y))$ converge to $f_1 \in \mathcal{C}(X, Y)$ and $f_2 \in \mathcal{C}(X, Y)$ respectively. For any filter $P \in \mathcal{F}(X)$ converging to $p \in X$, we have

$$\begin{aligned}(F_1 + F_2)(P) &= [\{(A_1 + A_2)(B) \mid A_i \in F_i, B \in P\}] \\ &\supset [\{A_1(B_1) + A_2(B_2) \mid A_i \in F_i, B_i \in P\}] \\ &= F_1(P) + F_2(P) \to f_1(p) + f_2(p) = (f_1 + f_2)(p).\end{aligned}$$

The continuity of scalar multiplication is proven similarly. ∎

Lemma 1.3.3. *Let X, Y be topological spaces such that X is first countable. Then a sequence $\{f_k\}_{k\in\mathbb{N}} \subset \mathcal{C}(X, Y)$ converges to a function $f \in \mathcal{C}(X, Y)$ if and only if $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x$ in X.*

Proof: First, let F be the derived filter of $\{f_k\}_{k\in\mathbb{N}}$ and suppose that F converges to a function $f \in \mathcal{C}(X, Y)$. Consider a sequence $\{x_k\}_{k\in\mathbb{N}}$ converging to $x \in X$, with its derived filter P. Now let V be a neighborhood of $f(x)$ in Y. By the definition of continuous convergence, we have $F(P) \to f(x)$, and so $V \in F(P)$. This means that V contains an element of the base of $F(P)$, namely a set

$$B_{n_0,m_0} = \{f_n(x_m) \mid n \geq n_0,\ m \geq m_0\}.$$

Hence, for $k \geq \max(n_0, m_0)$, we have $f_k(x_k) \in V$, as desired.

Conversely, suppose that $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x \in X$. First let us show that for any subsequence $\left\{f_{n_k}\right\}_{k\in\mathbb{N}}$, we have $f_{n_k}(x_k) \xrightarrow[k\to\infty]{} f(x)$.
To this end, let $n \in \mathbb{N}$. We let $z_n = x_{\varphi(n)}$, where $\varphi(n)$ is the minimal number of those $k \in \mathbb{N}$ that satisfy $n \leq n_k$. We clearly see that $n_1 \geq n_2$ implies $\varphi(n_1) \geq \varphi(n_2)$ and that $\varphi(n_k) = k$ for all $k \in \mathbb{N}$. Therefore, the sequence $\{z_n\}_{n\in\mathbb{N}}$ converges to x. Hence we have $f_k(z_k) \xrightarrow[k\to\infty]{} f(x)$, and so

$$f_{n_k}(x_k) = f_{n_k}\left(z_{n_k}\right) \xrightarrow[k\to\infty]{} f(x). \tag{1}$$

Now, to show that $f_k \to f$ in $\mathcal{C}(X, Y)$, we need to show that the derived filter F generated by the sets $C_n = \{f_k \mid k \geq n\}$ converges to f. To this end, let P be a filter in

X converging to a point $x \in X$. Note that since X is first countable, the filter P has a countable base $\{A_m\}_{m \in \mathbb{N}}$ of neighborhoods of x. To show that $F(P)$ converges to $f(x)$, assume the contrary. Then some neighborhood V of $f(x)$ does not lie in $F(P)$, which is to say that for all $n, m \in \mathbb{N}$, we have $C_n(A_m) \not\subset V$. That means that for all $n, m \in \mathbb{N}$, there are $k_m \geq m$ and $x_m \in A_m$ such that $f_{k_n}(x_m) \notin V$.
In particular, we have $f_{k_n}(x_n) \notin V$ for all $n \in \mathbb{N}$. At the same time, since the sets A_n generate P, we see that $x_n \xrightarrow[n \to \infty]{} x$, so we must have $f_{k_n}(x_n) \xrightarrow[n \to \infty]{} f(x)$ by (1), a contradiction. ■

2. The spaces of curried functions

2.1. General functions

Definition 2.1.1. Let $n \in \mathbb{N}_0 = \{0, 1, 2, ...\}$. The set $H_n(\mathbb{R})$ of *curried n-variable functions* is a topological vector space defined recursively as follows:
- If $n = 0$, we set $H_0(\mathbb{R}) = \mathbb{R}$;
- If $H_n(\mathbb{R})$ is defined, we let $H_{n+1}(\mathbb{R})$ to be the set of all functions from $\mathbb{R}$ to $H_n(\mathbb{R})$. This set is given the pointwise linear structure and the topology of pointwise convergence. Lemma 1.2.1 and Lemma 1.2.2 ensure that $H_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space.

Definition 2.1.2. Let $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ be a function. Define

$$\gamma\left(\tilde{f}\right)(x_1)(x_2)...(x_n) = \tilde{f}(x_1, x_2, ..., x_n).$$

Then $\gamma : \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}) \to H_n(\mathbb{R})$ is clearly a bijection. The function $f = \gamma\left(\tilde{f}\right)$ will be called the *curried version* of $\tilde{f}$.

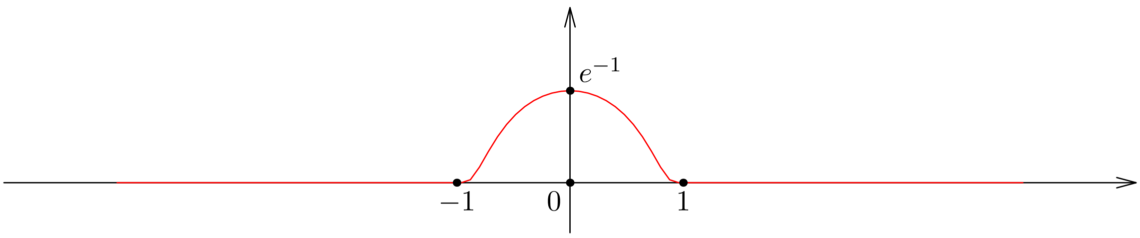
2.2. Continuous functions

Definition 2.2.1. Let $n \in \mathbb{N}_0$. The set $C_n(\mathbb{R})$ of *curried continuous functions* is a Hausdorff topological vector space defined recursively as follows:
- If $n = 0$, we set $C_0(\mathbb{R}) = \mathbb{R}$.
- If $C_n(\mathbb{R})$ is defined, we set $C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$, endowed with pointwise linear structure and the compact-open topology. Lemma 1.3.1 and Lemma 1.3.2 ensure that $C_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space, provided that $C_n(\mathbb{R})$ is.

Example 2.2.1. The topology on $C_n(\mathbb{R})$ is finer than the topology induced from $H_n(\mathbb{R})$, and does not coincide with it, unless $n = 0$. To see this, take $n = 1$ and consider the classical function

$$\varphi(x) = \begin{cases} \exp\left(\frac{1}{|x|^2 - 1}\right), & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

This is a $C^\infty(\mathbb{R})$ function supported on $[-1, 1]$. Its graph looks as follows:



Now consider the following sequence of functions:

$$f_k(x) = e \cdot \varphi\Big(k \cdot \Big(x - \frac{1}{k}\Big)\Big).$$

Clearly, f_k is infinitely differentiable on $\mathbb{R}$, supported on $[0, 2/k]$, and has a maximum at $x = 1/k$ with $f_k(1/k) = 1$. It is clear that for all $x \in \mathbb{R}$, we have $f_k(x) \xrightarrow[k\to\infty]{} 0$, so $f_k \to 0$ in the topology induced from $H_1(\mathbb{R})$. However, f_k do not converge to 0 in the native topology of $C_1(\mathbb{R})$. To see this, take $x_k = 1/k$ and observe that $f_k(x_k) \equiv 1 \neq 0$.

A similar argument applies for any other $n \geq 2$. Hence we conclude that $C_n(\mathbb{R})$ is not a subspace of $H_n(\mathbb{R})$.

Proposition 2.2.1. *Let $n \in \mathbb{N}_0$. Then the map*

$$\gamma : \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \to C_n(\mathbb{R})$$

defined by $\gamma(f)(x_1)(x_2)...(x_n) = f(x_1, ..., x_n)$, is well-defined and is a homeomorphism.

Proof: If $n = 0$, the statement is trivial. Assume it is proven for $C_n(\mathbb{R})$. By Proposition 1.3.2, we have

$$\mathcal{C}(\mathbb{R}^{n+1}, \mathbb{R}) = \mathcal{C}(\mathbb{R} \times \mathbb{R}^n, \mathbb{R}) \cong \mathcal{C}(\mathbb{R}, \mathcal{C}(\mathbb{R}^n, \mathbb{R})) \cong \mathcal{C}(\mathbb{R}, C_n(\mathbb{R})) = C_{n+1}(\mathbb{R}),$$

q.e.d. ■

2.3. Differentiable functions

Definition 2.3.1. Let $n \in \mathbb{N}_0$. The set $D_n(\mathbb{R})$ of *curried differentiable functions* is a topological vector space defined recursively as follows:

- If $n = 0$, we set $D_0(\mathbb{R}) = \mathbb{R}$, as usual;
- If $D_n(\mathbb{R})$ is defined for some n, we let $D_{n+1}(\mathbb{R})$ consist of all functions $f \in \mathcal{C}(\mathbb{R}, D_n(\mathbb{R}))$ such that there is a function $f' \in C_{n+1}(\mathbb{R})$ (called the derivative of f) which satisfies

$$\frac{f(x+h) - f(x)}{h} \xrightarrow[h\to 0]{} f'(x) \in H_n(\mathbb{R})$$

for all $x \in \mathbb{R}$, where **the convergence is taken in the space $H_n(\mathbb{R})$**. Clearly, if $f, g \in D_{n+1}(\mathbb{R})$ and $\alpha, \beta \in \mathbb{R}$, we have

$$(\alpha f + \beta g)' = \alpha f' + \beta g',$$

so in particular, $\alpha f + \beta g \in D_{n+1}(\mathbb{R})$.
We endow $D_{n+1}(\mathbb{R})$ with the coarsest topology in which the maps

$$\begin{aligned} i : D_{n+1}(\mathbb{R}) &\to \mathcal{C}(\mathbb{R}, D_n(\mathbb{R})), \\ f &\mapsto f \end{aligned} \qquad \begin{aligned} d : D_{n+1}(\mathbb{R}) &\to C_{n+1}(\mathbb{R}) \\ f &\mapsto f' \end{aligned} \qquad (2)$$

are continuous. This topology will be called the *differential topology*. We prove below that $D_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space.

Remark 2.3.1. It is clear that the inclusion map $i : D_n(\mathbb{R}) \to C_n(\mathbb{R})$ is continuous, meaning that the topology on $D_n(\mathbb{R})$ is finer than the topology induced from $C_n(\mathbb{R})$.

Lemma 2.3.1. *For all $n \in \mathbb{N}_0$, the topological space $D_n(\mathbb{R})$ is Hausdorff.*

Proof: Since $C_n(\mathbb{R})$ is Hausdorff and $i : D_n(\mathbb{R}) \to C_n(\mathbb{R})$ is continuous and injective, we see that $D_n(\mathbb{R})$ is also Hausdorff. ■

Lemma 2.3.2. *For all $n \in \mathbb{N}_0$, the linear and topological structures on $D_n(\mathbb{R})$ are compatible, making $D_n(\mathbb{R})$ a topological vector space.*

Proof: We prove by induction over n. If $n = 0$, we have $D_0(\mathbb{R}) = \mathbb{R}$ with the usual topology and linear structure. Suppose now that $D_{n-1}(\mathbb{R})$ is a TVS. We need to establish the continuity of the following two maps:

$$+ : D_n(\mathbb{R}) \times D_n(\mathbb{R}) \to D_n(\mathbb{R}), \qquad\qquad \cdot : \mathbb{R} \times D_n(\mathbb{R}) \to D_n(\mathbb{R}).$$

In turn, by the definition of the differential topology, this task is equivalent to proving that the maps $i \circ (+), d \circ (+), i \circ (\cdot), d \circ (\cdot)$ are continuous, where the maps i and d are defined in (2).

To this end, note that we have commutative diagrams

$$\begin{tikzcd}
D_n(\mathbb{R}) \times D_n(\mathbb{R}) \arrow[r, "(+)"] \arrow[d, hook, "i \times i"'] & D_n(\mathbb{R}) \arrow[d, hook, "i"] \\
\mathcal{C}(\mathbb{R}, D_{n-1}(\mathbb{R})) \times \mathcal{C}(\mathbb{R}, D_{n-1}(\mathbb{R})) \arrow[r, "(+)"'] & \mathcal{C}(\mathbb{R}, D_{n-1}(\mathbb{R}))
\end{tikzcd}$$

$$\begin{tikzcd}
D_n(\mathbb{R}) \times D_n(\mathbb{R}) \arrow[r, "(+)"] \arrow[d, "d \times d"'] & D_n(\mathbb{R}) \arrow[d, "d"] \\
C_n(\mathbb{R}) \times C_n(\mathbb{R}) \arrow[r, "(+)"'] & C_n(\mathbb{R})
\end{tikzcd}$$

$$\begin{tikzcd}
\mathbb{R} \times D_n(\mathbb{R}) \arrow[r, "(\cdot)"] \arrow[d, hook, "\mathrm{id} \times i"'] & D_n(\mathbb{R}) \arrow[d, hook, "i"] \\
\mathbb{R} \times \mathcal{C}(\mathbb{R}, D_{n-1}(\mathbb{R})) \arrow[r, "(\cdot)"'] & \mathcal{C}(\mathbb{R}, D_{n-1}(\mathbb{R}))
\end{tikzcd}$$

$$\begin{tikzcd}
\mathbb{R} \times D_n(\mathbb{R}) \arrow[r, "(\cdot)"] \arrow[d, "\mathrm{id} \times d"'] & D_n(\mathbb{R}) \arrow[d, "d"] \\
\mathbb{R} \times C_n(\mathbb{R}) \arrow[r, "(\cdot)"'] & C_n(\mathbb{R})
\end{tikzcd}$$

Since the down-then-right path in each diagram is continuous, the right-then-down paths are continuous as well, and we are done. ∎

Theorem 2.3.1. *Let $n \in \mathbb{N}$. A curried function $f \in H_n(\mathbb{R})$ lies in the class $D_n(\mathbb{R})$ if and only if its counterpart $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ has continuous partial derivatives on $\mathbb{R}^n$. Moreover, for each $1 \leq i \leq n$, we have*

$$\frac{\partial \tilde{f}}{\partial x_i}(x_1, x_2, ..., x_n) = f(x_1)...(x_{i-1})'(x_i)...(x_n). \tag{3}$$

Proof: We employ induction over $n \in \mathbb{N}_0$. If $n = 1$, the statement clearly holds. Now, consider $n \geq 2$ and assume that the statement holds for $n - 1$.
First, let $f \in D_n(\mathbb{R})$ and $(x_1, x_2, ..., x_n) \in \mathbb{R}^n$. For $i \geq 2$, we have

$$\begin{aligned}\frac{\partial \tilde{f}}{\partial x_i}(x_1, ..., x_n) &= \lim_{h \to 0} \frac{\tilde{f}(x_1, ..., x_i + h, ..., x_n) - \tilde{f}(x_1, ..., x_n)}{h} \\ &= \lim_{h \to 0} \frac{\widetilde{f(x_1)}(x_2, ..., x_i + h, ..., x_n) - \widetilde{f(x_1)}(x_2, ..., x_n)}{h} \\ &= \frac{\partial \widetilde{f(x_1)}}{\partial x_i}(x_2, ..., x_n) = f(x_1)...(x_{i-1})'(x_i)...(x_n)\end{aligned}$$

by the induction hypothesis, since $f(x_1) \in D_{n-1}(\mathbb{R})$. For the derivative with respect to x_1, we have

$$\begin{aligned}f'(x_1)(x_2)...(x_n) &= \left(\lim_{h \to 0} \frac{f(x_1 + h) - f(x_1)}{h}\right)(x_2)...(x_n) \\ &= \lim_{h \to 0} \frac{f(x_1 + h)(x_2)...(x_n) - f(x_1)...(x_n)}{h} \\ &= \lim_{h \to 0} \frac{\tilde{f}(x_1 + h, x_2, ..., x_n) - \tilde{f}(x_1, ..., x_n)}{h} = \frac{\partial \tilde{f}}{\partial x_1}(x_1, ..., x_n),\end{aligned}$$

so $\partial \tilde{f}/\partial x_1 = f'$, and we see that (3) holds. We also immediately observe that $\partial \tilde{f}/\partial x_1$ is continuous by Proposition 2.2.1, since $f' \in C_n(\mathbb{R})$. It remains to show that the derivatives with respect to $x_2, ..., x_n$ are continuous. To this end, let $(h_1, ..., h_n) \to 0 \in \mathbb{R}^n$. Since $D_n(\mathbb{R}) \subset \mathcal{C}(\mathbb{R}, D_{n-1}(\mathbb{R}))$, the function $f : \mathbb{R} \to D_{n-1}(\mathbb{R})$ is continuous, and so we have the convergence

$$f(x_1 + h_1) \xrightarrow[h_1 \to 0]{} f(x_1) \in D_{n-1}(\mathbb{R}).$$

Now, since the map $d : D_{n-1}(\mathbb{R}) \to C_{n-1}(\mathbb{R})$ is continuous, the above implies

$$f(x_1 + h_1)' \xrightarrow[h_1 \to 0]{} f(x_1)' \in C_{n-1}(\mathbb{R}).$$

From here, by Lemma 1.3.3, we have

$$\begin{aligned}&\frac{\partial \tilde{f}}{\partial x_2}(x_1 + h_1, ..., x_n + h_n) \\ &= f(x_1 + h_1)'(x_2 + h_2)...(x_n + h_n) \xrightarrow[h_1, h_2, ... h_n \to 0]{} f(x_1)'(x_2)...(x_n) = \frac{\partial \tilde{f}}{\partial x_2}(x_1, ..., x_n).\end{aligned}$$

A similar argument shows that the partial derivatives with respect to $x_3, ..., x_n$ are continuous as well.

Now, we prove the other direction. Consider $f \in H_n(\mathbb{R})$ such that $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ has continuous partial derivatives. To show that $f \in D_n(\mathbb{R})$, we follow three steps:

1. For all $x_1 \in \mathbb{R}$, we have $f(x_1) \in D_{n-1}(\mathbb{R})$. This holds by the induction hypothesis, since $\widetilde{f(x_1)} : \mathbb{R}^{n-1} \to \mathbb{R}$ clearly has continuous partial derivatives, being a restriction of $\tilde{f}$.
2. The function $f : \mathbb{R} \to D_{n-1}(\mathbb{R})$ is continuous. To show this, consider a sequence $\left\{x_k^{(1)}\right\}_{k \in \mathbb{N}} \subset \mathbb{R}$ converging to $x_0^{(1)} \in \mathbb{R}$. We aim to prove that $f\left(x_k^{(1)}\right) \to f\left(x_0^{(1)}\right)$ in $D_{n-1}(\mathbb{R})$. This is equivalent to showing that
$$f\left(x_k^{(1)}\right) \to f\left(x_k^{(1)}\right) \in \mathcal{C}(\mathbb{R}, D_{n-2}(\mathbb{R})) \quad \text{and} \quad f\left(x_k^{(1)}\right)' \to f\left(x_0^{(1)}\right)' \in C_{n-1}(\mathbb{R}). \tag{4}$$
The latter of these conditions holds by the induction hypothesis, since
$$\frac{\partial \tilde{f}}{\partial x_2}(x_1, ..., x_n) = \frac{\partial \widetilde{f(x_1)}}{\partial x_2}(x_2, ..., x_n) \underset{\text{by (3)}}{=} f(x_1)'(x_2)...(x_n), \tag{5}$$
and $\partial \tilde{f} / \partial x_2$ is continuous, which means that the map $x \mapsto f(x)'$ is continuous. To prove the former condition in (4), we again consider a sequence $x_k^{(2)} \to x_0^{(2)} \in \mathbb{R}$ and aim to prove
$$f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right) \in D_{n-2}(\mathbb{R}),$$
which (unless $n = 2$) is again equivalent to
$$f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \to f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right) \quad \text{and} \quad f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right)' \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right)'.$$
The latter follows from the induction hypothesis and the continuity of $\partial \tilde{f} / \partial x_3$, similarly to (5), and the former can again be expanded further. This chain leads us finally to the convergence
$$f\left(x_k^{(1)}\right)\left(x_k^{(2)}\right)...\left(x_k^{(n)}\right) \to f\left(x_0^{(1)}\right)\left(x_0^{(2)}\right)...\left(x_0^{(n)}\right) \in \mathbb{R},$$
which is evident since $f \in C_n(\mathbb{R})$ by Proposition 2.2.1. Hence we see that f lies in $\mathcal{C}(\mathbb{R}, D_{n-1}(\mathbb{R}))$. We have also established (3) for f and $i \geq 2$, via (5).
3. Finally, we need to find $f' \in C_n(\mathbb{R})$ such that
$$\frac{f(x+h) - f(x)}{h} \underset{h \to 0}{\longrightarrow} f'(x) \in H_{n-1}(\mathbb{R}) \tag{6}$$
for all $x \in \mathbb{R}$. Indeed, following (3), let us define $f' := \gamma\left(\partial \tilde{f} / \partial x_1\right)$. We immediately see by (Proposition 2.2.1) that f' so defined lies in $C_n(\mathbb{R})$. To see that (6) holds, consider $x_1, ..., x_n \in \mathbb{R}$ and write

$$
\begin{aligned}
f'(x_1)(x_2)...(x_n) &= (\partial \frac{\tilde{f}}{\partial x_1}(x_1, ..., x_n) \\
&= \lim_{h \to 0} \frac{\tilde{f}(x_1 + h, x_2, ..., x_n) - \tilde{f}(x_1, ..., x_n)}{h} \\
&= \left(\lim_{h \to 0} \frac{f(x_1 + h) - f(x_1)}{h} \right) (x_2)...(x_n),
\end{aligned}
$$

and we are done.

■

Bibliography

[1] N. Bourbaki, *General Topology, Part I*. 1966.

[2] N. Bourbaki, *General Topology, Part II*. 1966.

[3] R. Beattie and H.-P. Butzmann, *Convergence Structures and Applications to Functional Analysis*. 2002.